

Practical – Development/Programming Assessment

Course: Fullstack AI – Batch - 2

You have to program/Develop : 1 question from Section A and 1 question from section B

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Section A: Machine Learning [Marks 30]

Regression....? Open to any ML

Dataset:

Kindly see: "RealEstate-dataset.csv"

Konya Real Estate Dataset (17.07.2024)

Column	Value	Description
property_status	satilik	For sale
property_category	konut	Residential
property_subcategory	daire	Apartment
city	konya	Specific city
town	meram, selcuklu, karatay	Specific town
neighborhood	string	Specific neighborhood
district	string	District information
floor_count	integer between 1 and 18	Number of floors
floor	integer between 0 and 15	Floor number
building_age	0, 1, 2, 3, 4, 5 - 10, 11 - 15, 16 - 20, 21 +	Age of the building
heating_type	kombi dogalgaz, yerden isitma, merkezi pay olcer, merkezi dogalgaz, soba	Heating type
gross_area_m2	integer	Gross area in square meters
net_area_m2	integer	Net area in square meters
room_count	1+1, 2+1, 3+1, 4+1, 5+1, 6+1	Number of rooms
bathroom_count	integer between 1 and 2	Number of bathrooms
price	integer	Price

Or

Classification

See File attached with name “loan_data.csv”

About Dataset

About the data and what to do...

publicly available data from LendingClubzxxx.com. Lending Club connects people who need money (borrowers) with people who have money (investors). Hopefully, **as an investor you would want to invest in people who showed a profile of having a high probability of paying you back.**

We will use lending data from 2007-2010 and be trying to classify and predict whether or not the borrower paid back their loan in full.

Here are what the columns represent:

credit.policy: 1 if the customer meets the credit underwriting criteria of LendingClub.com, and 0 otherwise.

purpose: The purpose of the loan (takes values "credit_card", "debt_consolidation", "educational", "major_purchase", "small_business", and "all_other").

int.rate: The interest rate of the loan, as a proportion (a rate of 11% would be stored as 0.11). Borrowers judged by LendingClub.com to be more risky are assigned higher interest rates.

installment: The monthly installments owed by the borrower if the loan is funded.

log.annual.inc: The natural log of the self-reported annual income of the borrower.

dti: The debt-to-income ratio of the borrower (amount of debt divided by annual income).

fico: The FICO credit score of the borrower.

days.with.cr.line: The number of days the borrower has had a credit line.

revol.bal: The borrower's revolving balance (amount unpaid at the end of the credit card billing cycle).

revol.util: The borrower's revolving line utilization rate (the amount of the credit line used relative to total credit available).

inq.last.6mths: The borrower's number of inquiries by creditors in the last 6 months.

delinq.2yrs: The number of times the borrower had been 30+ days past due on a payment in the past 2 years.

pub.rec: The borrower's number of derogatory public records (bankruptcy filings, tax liens, or judgments).

Steps - TO Do: [Neat, Clean, well-documented code]

Loading in DataFrame, Perform “Descriptive Statistics” Max-Operations, Input data analysis based on seaborn, Performing core operation of SK Learn, Perform metric analysis

[10 Marks for exceptional creative deep data analytical skills demonstration. For standard implementation – these marks will NOT be given.]

Section B: Deep Learning [Marks 30]

Time SERIES - LSTM or GRU

Steps - TO Do:[Neat, Clean, well-documented code]

Design Model with related layers, Performing core operation of Tensflow-Keras based deep learning, Perform Model metric analysis

[10 Marks for exceptional creative deep data analytical skills demonstration. For standard implementation – these marks will NOT be given.]

Dataset

Timeseries Trending YouTube Videos: 2019-04-15 to 2020-04-15

This dataset is a csv of one of the archived historical database tables queried from my non public database that contains time series data for period of 2019-04-15 to 2020-04-15. Video data was captured from the time they first appeared on trending list, and TSD exists until the video is removed from trending list.

This snapshot contains data for the 11,369 videos that appeared on trending within the timeframe, with 1,541,128 records total TSD.

TSD in this dataset was spidered on variable frequency at the start, however it should stabilize to every 30 minutes later in the dataset.

Data provided in this dataset is:

ytvideoid (the id of the video according to Youtube), views, comments, likes, dislikes (This is prior to the removal of dislikes as publicly viewable data)

See File attached with name “videostats.csv”